34 BIOPHYSICS OF SOLUTIONS *Pr B. Boutabia-Chéraitia*

↬ J_d can also be written this way:

$$J_d = -D_m \frac{dC}{dx}$$

$D_m = D(1-\sigma)\frac{S_p}{S}$: diffusion coefficient of the solute in the membrane.

S_p : total pore area (permeable and impermeable).

↬ The **initial diffusive flux** is written:

$$J_{d_i} = \frac{Q_i}{S} = \frac{D}{l}\frac{S'}{S}|\Delta C_i| \equiv P|\Delta C_i|$$

$$P = \frac{D}{l}\frac{S'}{S} = \frac{D_m}{l} \qquad [P] = m/s$$

P : **Diffusive permeability** per unit membrane area.

Remark

$$D = \frac{kT}{f}$$

$k = 1.38 \times 10^{-23} J/K$: ***Boltzmann***'s constant.

f : **Friction coefficient** of the solvent on the diffusing particles.

35 BIOPHYSICS OF SOLUTIONS *Pr B. Boutabia-Chéraitia*

$f = 6\pi\eta r$: for spherical particles of radius r.

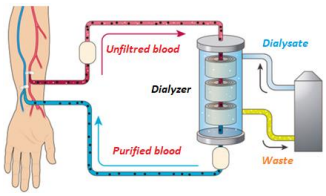
η : **viscosity** coefficient characterizing the resistance to flow.

$M = \mathcal{N}m = \mathcal{N}\rho\frac{4}{3}\pi r^3 \Rightarrow r = Cste\sqrt[3]{\frac{M}{\rho}}$ and $D = Cste\frac{T}{\eta}\sqrt[3]{\frac{\rho}{M}}$

Medical application: dialysis and hemodialysis

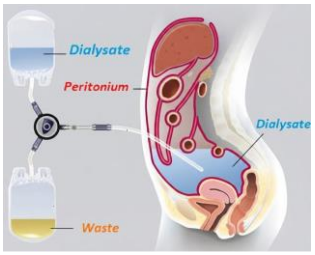
Dialysis is a method for **purifying** solutions. In medicine, it purifies the blood of its waste products (urea, creatinine, excess potassium, or liquid, etc.) across a membrane when the kidneys can no longer do so.

Hemodialysis is a method allowing exchanges **outside** the body between the blood and a fluid called **dialysate**, through an artificial filter called a **dialyzer**.



36 BIOPHYSICS OF SOLUTIONS *Pr B. Boutabia-Chéraitia*

Peritoneal dialysis is a method allowing exchanges **inside** the body between the blood and the dialysate, across the **peritoneum**, a richly vascularized membrane that envelops the organs of the abdominal cavity and the digestive tract.



Exercise 9

Two aqueous solutions of the same substance, with concentrations $20mmol/l$ and $90mmol/l$, are separated par dialysis membrane with a total area of $3cm^2$. The pores area represents 1%, and the thickeness is $10\mu m$. Given that the initial molar flow rate of the solute is $4.2pmol/s$, calculate the diffusive permeability of the membrane and the diffusion coefficient.

Solutions:

$$\left.\begin{array}{l} \hookrightarrow J_{d_i} = \frac{Q_i}{S} \\ J_{d_i} = P|\Delta C_i| \end{array}\right\} \Rightarrow P = \frac{Q_i}{S|\Delta C_i|} = \frac{4.2\times10^{-12}}{3\times10^{-4}\times70} = 0.2\times10^{-9} m/s$$

37 BIOPHYSICS OF SOLUTIONS *Pr B. Boutabia-Chéraitia*

$$\hookrightarrow P = \frac{D_m}{l} = \frac{D(1-\sigma)}{l}\frac{S_p}{S}$$

$$\sigma = 0 \Rightarrow P = \frac{D}{l}\frac{S_p}{S} \qquad S_p = 10^{-2}S$$

$$\Rightarrow D = \frac{P|S}{S_p} = \frac{0.2\times10^{-9}\times10^{-5}}{10^{-2}} = 0.2\times10^{-12} m^2/s$$

Exercise 10

Two aqueous solutions of sucrose, with concentrations of $0.3mol/l$ and $0.1mol/l$, are separated by a dialysis membrane with a surface area of $20cm^2$ and a thickness of $0.2mm$. Given that the diffusive permeability of the membrane to sucrose is $2.5\times10^{-6}m/s$, calculate the initial molar flow rate in $\mu mol/s$.

Solutions:

$$Q_i = PS|\Delta C_i| = 2.5\times10^{-6}\times20\times10^{-4}(0.3-0.1)\times10^3$$

$$= 10^{-6} mol/s = 1\mu mol/s$$

38 BIOPHYSICS OF SOLUTIONS *Pr B. Boutabia-Chéraitia*

Exercise 11

A myoglobin molecule $(17kg/mol)$ modeled as a sphere, has a diffusion coefficient of $D = 1.2 \times 10^{-10} m^2/s$ in water at $0°C$, the viscosity of water being $10^{-3} Pa.s$.

1) Calculate its radius in (nm)
2) Deduce the molar mass of a spherical macromolecule having the same density as myoglobin, given that at $37°C$ its diffusion coefficient in a solvent with a viscosity of $1mPa.s$ is $0.65m^2/s$.

Solutions:

1) $D = \frac{kT}{6\pi\eta r} \Rightarrow r = \frac{kT}{6\pi\eta D} = \frac{1.38\times10^{-23}\times273}{6\pi\times10^{-3}\times1.2\times10^{-10}} = 1.66\times10^{-9}m = 1.66nm$

2) Myoglobine $\rightarrow D_1 = Cste\frac{T_1}{\eta_1}\left(\frac{\rho_1}{M_1}\right)^{1/3}$

Macromolecule $\rightarrow D_2 = Cste\frac{T_2}{\eta_2}\left(\frac{\rho_2}{M_2}\right)^{1/3}$

$\frac{D_1}{D_2} = \frac{T_1}{T_2}\times\frac{\eta_2}{\eta_1}\times\left(\frac{\rho_1}{\rho_2}\right)^{1/3}\times\left(\frac{M_2}{M_1}\right)^{1/3} = \frac{T_1}{T_2}\times\left(\frac{M_2}{M_1}\right)^{1/3}$

$\eta_1 = \eta_2 \quad \rho_1 = \rho_2$

39 BIOPHYSICS OF SOLUTIONS *Pr B. Boutabia-Chéraitia*

$\Rightarrow M_2 = M_1\times\left(\frac{D_1}{D_2}\times\frac{T_2}{T_1}\right)^3 = 4880g/mol$

Exercise 12:

An artificial kidney used in chronic hemodialysis has a diffusion surface area of $S = 0.5m^2$ and an average pore length of $l = 50\mu m$. Across this surface, the patient's blood is brought into equilibrium with a constantly renewed fluid (dialysate) that is isotonic to plasma and free of urea. The diffusion coefficient of urea in the membrane is $10^{-9}m^2/s$. If a dialysis patient has an initial blood urea concentration of $5g/l$, calculate the initial mass flow rate of urea in (mg/s).

Solutions:

Blood	Dialysate
$C_{1i} = C_0$	$C_{2i} = 0$
V_0	V_e
$m_0 = C_0V_0$	
①	②

$Q_i = \left(\frac{dm}{dt}\right)_i = PS|\Delta C_i| = \frac{D_mS}{l}|\Delta C_i|$

$|\Delta C_i| = C_0 - 0 = C_0$

$\Rightarrow Q_i = \frac{D_mS}{l}C_0$

$= \frac{10^{-9}\times0.5}{50\times10^{-6}}\times5\times10^3 = 5\times10^{-2}g/s = 50mg/s$